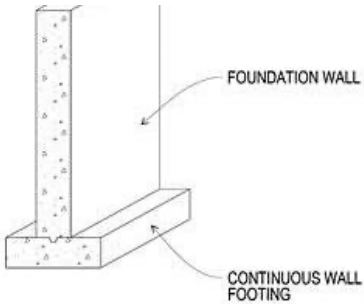
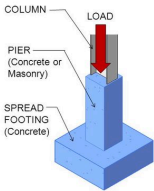
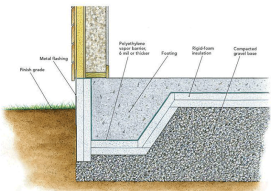
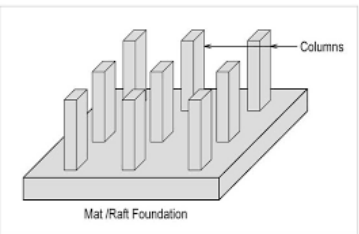
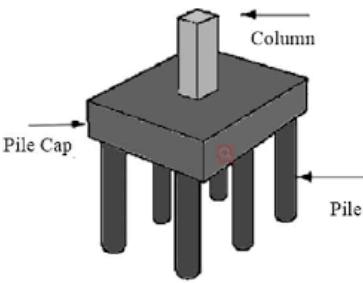
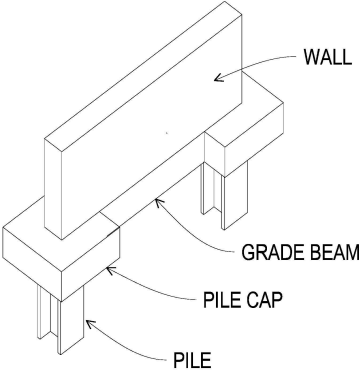


1. Research the following foundation types. Document each type with a sketch and describe the conditions under which each might be used in a table similar to the table shown.

Typical Foundation Systems	Usage	Diagram/Sketch
Continuous (Strip) Footing with Foundation Wall	Builders use strip footings to support load-bearing walls. They work well for residential homes with basements in stable soil.	
Spread Footing with Pier	Engineers place spread footings under individual columns. They distribute heavy point loads across a broader soil area.	Spread (Column) Footing  • The widened part of a foundation that spreads a column load over a broader area of soil • Design based on – Soil Bearing Capacity – Column Load
Slab-on-Grade with Thickened Slab	Contractors pour these shallow foundations directly on the ground. They thickened edges supported exterior walls in warm climates.	
Mat (Floating) Foundation	A mat foundation supports the entire structure on one thick concrete slab. Builders pick this for weak soils.	

Typical Foundation Systems	Usage	Diagram/Sketch
Pile and Pile Cap	Pile transfers heavy building loads deep into the earth. The cap groups multiple piles to support a column.	
Cast-in-Situ Pile and Grade Beam	Workers drill holes and pour concrete on site. The grade beam connects the piles to support the walls.	

2. The Keystone Library Renovation site is in **Zone A** (specifically Zone AE). Consequently, the lowest floor must be elevated to or above the Base Floor Elevation (BFE). Any enclosed areas below the BFE must be equipped with flood openings or vents. All construction material used below the BFE must be flood-resistant. Electrical, heating, ventilation, plumbing, and air conditioning equipment must all be elevated above the BFE.